Relative effects of seagrass canopy, wave coupling and bed mobility on Lagrangian transport in a shallow lagoon

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Abstract

Coastal lagoon models are routinely validated against water level, although connectivity and dispersal depend on transport. This study asks which modelled processes control transport in a shallow vegetated lagoon and whether water level identifies them. A three-dimensional Delft3D Flexible Mesh model of the Stagnone di Marsala, a micro-tidal Sicilian lagoon of sub-metre mean depth colonised by *Posidonia oceanica*, was coupled to SWAN over nine days of July 2025. Eight configurations form a complete factorial in wave coupling, bed mobility and seagrass canopy, the canopy parameterised from published measurements and mapped by satellite, and each was scored against three tide gauges and 35 surface drifters withheld from model decisions. On the fixed bed, the canopy raises Liu–Weisberg skill by 0.120 and 0.114, both bootstrap intervals clear of zero, against -0.022 and -0.028 for wave coupling. Water level separates the canopy treatments only at the gauge inside the vegetated basin, by 26 and 27 mm in anomaly RMSE against at most 4 mm near the inlets. With waves, the uncalibrated mobile bed brings a bare configuration to the skill of the vegetated ones through bathymetry reshaped by implausible erosion, which the interior gauge exposes and the drifters do not. The canopy is therefore the first-order control on transport here. These results rest on one nine-day window and a stratified initial state. In shallow meadows of this kind, a vertically resolved canopy drag is likely to be needed to reproduce surface transport, even where water level near the inlets is matched without it.

Keywords: Coastal lagoon, Delft3D Flexible Mesh, Canopy drag, *Posidonia oceanica*, Lagrangian drifters, Three-dimensional circulation

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1. Introduction

Coastal lagoons are among the most productive coastal environments on Earth, concentrating fisheries and biodiversity along roughly 13% of the global shoreline (Carrasco et al., 2016; Kjerfve, 1994). The shallow, semi-enclosed geometry that sustains this productivity also makes them acutely sensitive to the atmospheric and tidal forcing that drives water exchange and renewal. In the Mediterranean, where tidal amplitudes rarely exceed 0.3 m, the balance among tide, wind stress and seiches varies from one lagoon to another, but wind consistently emerges as a significant driver of horizontal exchange, and through it of the residence times and salinity gradients that condition benthic habitat (Niedda and Greppi, 2007; Umgiesser et al., 2014). In many of these lagoons that habitat is seagrass, principally *Posidonia oceanica* (L.) Delile and *Cymodocea nodosa* (Ucria) Ascherson, and the relation runs in both directions, since the canopies that depend on the circulation also act on it, attenuating incident wave energy (Infantes et al., 2012) and extracting momentum from the mean flow through a drag that modifies near-bed turbulence and the vertical velocity structure (Nepf, 2012). These meadows are declining, global seagrass extent having fallen in all seven bioregions, by 19.1% of the surveyed area since 1880 (Waycott et al., 2009; Dunic et al., 2021), with the fastest recent losses where meadows are exposed to coastal development and degraded water quality (Turschwell et al., 2021). How circulation sets residence times, dispersal and habitat connectivity in vegetated lagoons is therefore a question for management as much as for science, and it concerns transport pathways, a quantity that water level, the usual validation target of lagoon models, does not measure directly.

The *Stagnone di Marsala*, on the north-western coast of Sicily, is a lagoon of this kind. Its mean depth is about 0.95 m, and its floor is largely covered by meadows of *P. oceanica* and *C. nodosa* (Section 2). Satellite monitoring over 2003–2024 has documented a 75% reduction in the extent of the *P. oceanica* atoll and cordon formations (Maltese et al., 2025), so the vegetated state a model of this system must represent is itself changing.

Numerical studies of the Stagnone have each been built around the question they addressed. La Loggia et al. (2004) and Di Marca et al. (2009) used depth-averaged models to characterise tidal flushing, residence times and turbidity, for which the basin-scale water budget is the

relevant quantity, and Ciraolo et al. (2009a) showed through particle tracking how strongly simulated trajectories depend on the velocity field they are advected through. De Marchis et al. (2012) resolved the vertical and showed that wind drives a recirculation with a near-bed return flow even at sub-metre depths, with the meadow represented through Nikuradse roughness heights assigned by seagrass class and the model validated against velocity and water level at several stations. Ingrassia et al. (2024) calibrated a depth-averaged model against water level for a range of vegetation scenarios, a setting in which a uniform equivalent roughness is the appropriate representation of the canopy. Building on these studies, three aspects of the problem remain open for this basin. The first is the representation of the canopy as a drag resolved over the vertical extent of the plants, a formulation that acts on the shape of the velocity profile as well as on its magnitude (Nepf, 2012) and that is now available in three-dimensional solvers. The second is the effect of waves, which in comparably shallow Mediterranean lagoons alter bed stress and mobilise sediment (Carniello et al., 2011; Morote-Sánchez et al., 2025), together with that of a bed evolving under their combined action with the currents. The third is the assessment of transport itself, the models having been evaluated mainly on water level and on currents at fixed stations. Whether a model that reproduces water level also reproduces transport, and which of the processes it carries are responsible when it does, therefore remains to be established for this basin, and the question extends to shallow vegetated lagoons in general, where Lagrangian observations are known to constrain a model differently from water level (Liu and Weisberg, 2011; Révelard et al., 2021).

The study takes up these three aspects by separating, in a three-dimensional model of the Stagnone, the contributions of the canopy, of wave coupling and of bed mobility, and by scoring every combination of the three against both water level and observed drifter trajectories (Section 3). It pursues three objectives. The first is to rank the three processes by their effect on Lagrangian transport skill. The second is to establish whether water level, the quantity against which shallow lagoon models are conventionally validated, can identify that ranking. The third is to measure how far the apparent effect of each process depends on how the other two are represented. The second and third are not specific to this basin, concerning what a class of validation data can certify and how a factorial behaves when a dominant factor is left out of it.

2. Study Area

The *Stagnone di Marsala* is located on the north-western coast of Sicily, sheltered from the western Sicilian channel by Isola Grande, an elongated barrier island running parallel to the mainland coast (Figure 1). The enclosed basin extends approximately 11 km in the north–south direction and 2.5 km east–west, enclosing a total surface area of roughly 2200 ha (De Marchis et al., 2012). Two geomorphologically distinct sub-basins are separated by a *Posidonia oceanica* barrier reef (La Loggia et al., 2004): the northern sub-basin, with a mean depth of approximately 1 m and extensive intertidal flats, and the deeper southern sub-basin, with a mean depth of approximately 2 m and a maximum of about 3 m near the main inlet. Three islands punctuate the interior, the Phoenician archaeological site of Mothia in the south-central basin and the smaller islands of Santa Maria and Scuola in the central reaches (Ciraolo et al., 2009b). Salt pans (*saline*) occupy both margins of the lagoon, with extensive historic saltworks on the western barrier island (Isola Grande) and additional pans along the eastern mainland shore in the area facing Mothia (Basso et al., 2008). They constitute a distinctive shallow intertidal habitat within the lagoon system.

Hydrodynamic exchange with the open sea occurs through two morphologically distinct inlets. The northern inlet (approximately 400 m wide, 0.30–0.40 m deep) is a highly restricted passage whose tidal prism is severely limited by its shallow sill. A historically dredged channel (20 m wide, 1 m deep) partially compensates this restriction (Ciraolo et al., 2009b). The wide southern inlet (approximately 2900 m wide, 1.0–1.5 m deep) provides the primary pathway for mass transport and water mixing between the lagoon and the open sea (Ciraolo et al., 2009b). The tidal regime is microtidal, with free-surface amplitudes of approximately 0.30 m dominated by semi-diurnal harmonics (La Loggia et al., 2004). In the absence of riverine input, weak tidal exchange combined with high summer evaporation drives a pronounced salinity gradient. Annual lagoon-wide salinity spans 33–46 psu (Basso et al., 2008), with peak summer values approaching 48 psu in the more confined northern sub-basin, contrasting with near-marine conditions near the southern inlet (Mancuso et al., 2023). Wind forcing dominates east–west circulation within the basin, while tidal exchange controls the north–south component (Ciraolo et al., 2009b; De Marchis et al., 2012).

The seagrass mosaic covers the majority of the lagoon floor. *Posidonia oceanica* occupies

the central and southern portions of the basin, forming both continuous meadows and characteristic atoll patterns (10–20 m in diameter) that grade into reef formations oriented along the north–south axis. *Cymodocea nodosa* prevails in the shallower northern reaches, with *Caulerpa prolifera* replacing *P. oceanica* locally where hydrodynamic and sediment conditions deteriorate (La Loggia et al., 2004; Ciraolo et al., 2009b). In Mediterranean meadows of this type, *P. oceanica* areal density reaches 1000–1200 plants m^{-2} (Ciraolo et al., 2009a). The spatial distribution of submerged vegetation was first mapped from airborne hyperspectral imagery by Ciraolo et al. (2006a). The roughness parametrisation in this study rests on an independent classification of PlanetScope imagery for the 2023 epoch, described in Section 3.4. The lagoon is designated both as a *Riserva Naturale Orientata* and as a Natura 2000 site, reflecting the conservation importance of these meadows. The extent of the atoll and cordon formations nevertheless fell by approximately 75% between 2003 and 2024 (Maltese et al., 2025).

The observational dataset used in this study was collected during July 2025 (Figure 1). Water level was recorded at three stations, BocaNord (BN) at the northern inlet, BocaSud (BS) near the southern inlet, and AltaVilaEst (AE) on the eastern flank of the central basin. Wind speed and direction and atmospheric pressure were measured at two stations, AE (which also carries a water-level gauge and a water temperature sensor) and Mulino, a dedicated meteorological station located near the salt pans in the central-northern area. Water levels at BN, BS, and AE serve as the primary validation targets for the hydrodynamic model (Section 3.5). Open-sea boundary conditions are prescribed from CMEMS Mediterranean physical reanalysis products (Section 3.3). Offshore model performance is assessed against the tidal gauge on Marettimo island (JRC TAD 658), approximately 35 km to the west, which provides a free-surface reference uninfluenced by lagoonal dynamics. GPS-tracked surface drifters were released in the lagoon interior over a sub-period coinciding with the simulation window, and their trajectories form the basis of the Lagrangian validation (Section 3.5).

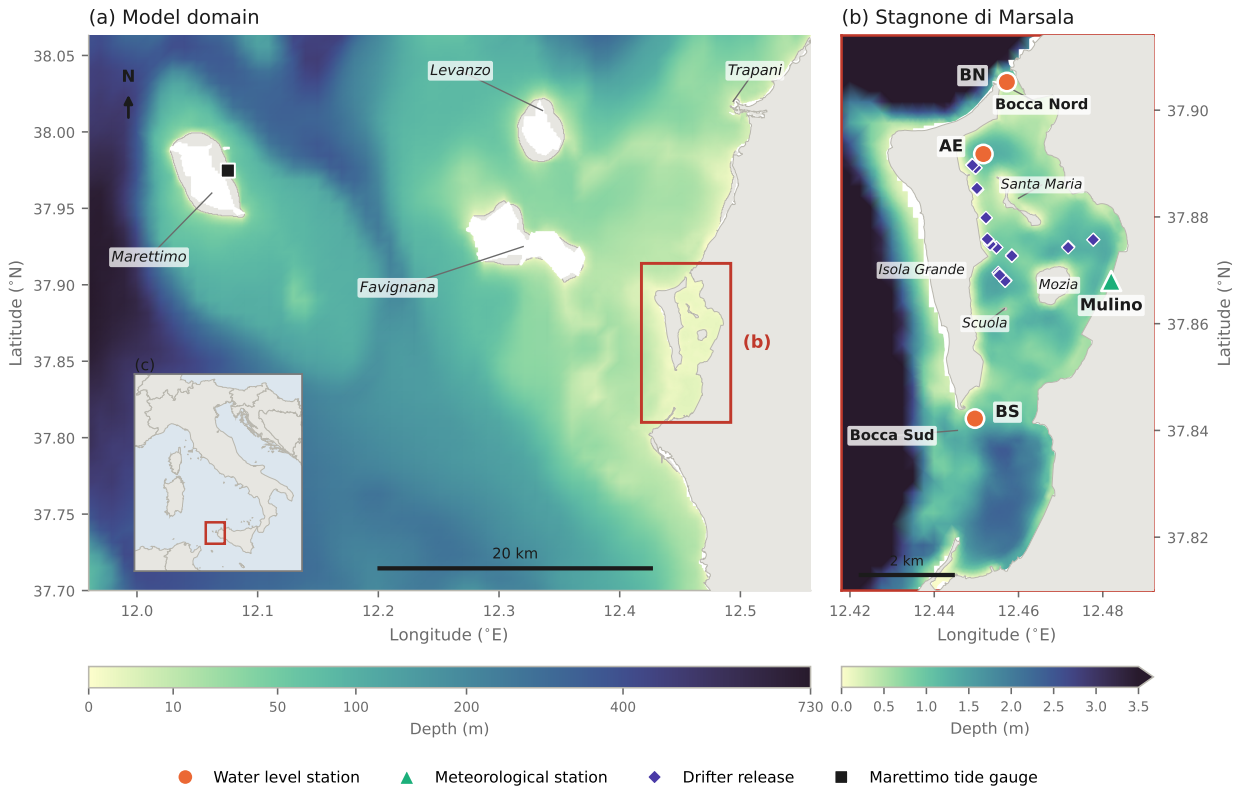


Figure 1: The study area. (a) The modelled domain, covering western Sicily, the Egadi archipelago and the adjacent shelf, with the Marettimo tide gauge used for offshore validation. The red box gives the extent of panel (b). (b) The Stagnone di Marsala, with the water level stations (BocaNord, AltaVilaEst, BocaSud), the Mulino meteorological station, and the 35 GPS drifter release positions of the twelve July 2025 deployments. (c) The domain within Italy. Colour gives depth below the reference level on the model mesh, on separate scales for (a) and (b).

3. Data and Methods

3.1. Hydrodynamic model

Circulation was simulated with D-Flow Flexible Mesh (D-Flow FM, kernel 1.2.184) from the Delft3D FM Suite 2026.01, an unstructured finite-volume solver for the three-dimensional shallow-water equations (Kernkamp et al., 2011). The mesh of 25,212 faces covers the lagoon together with an offshore extension reaching the western Sicilian channel, spanning 11.96–12.56°E and 37.70–38.06°N, and grades from about 880 m offshore to about 60 m in the lagoon and along the shorelines, so that about six cells span the 400 m northern inlet. The vertical carries 10 sigma layers of uniform thickness, about 0.095 m at the mean depth of 0.95 m, which resolves the wind-driven surface drift over the near-bed return flow reported by

De Marchis et al. (2012) and spreads the 0.40 m canopy of Section 3.4 over the lowest four layers and part of the fifth.

Turbulence closure used the k–ε model, and density followed the UNESCO equation of state with salinity and temperature as active constituents, so that the summer hypersaline gradient contributes baroclinically. Bed roughness was a uniform Manning coefficient of 0.023 s $m^{-1/3}$ in every member, the default of the solver, which stands for the bare substrate and was held uniform by design so that the canopy is the only spatially variable resistance in the ensemble. Wind drag followed the Charnock formulation with the coefficient taken from the ERA5 field. The simulation covered 1 to 10 July 2025 with an adaptive time step under a Courant limit of 0.7, capped at 30 s in the members coupled to the wave model and at 15 s in the uncoupled ones, on eight MPI partitions. The numerical settings, their grounds and the run times are given in the Supplementary Material, Section S1.

3.2. Wave model and coupling

Waves were computed with SWAN (Booij et al., 1999) through the Delft3D-WAVE module, on an outer grid of 69 × 54 points at about 800 m over the whole domain and an inner grid at about 90 by 110 m over the lagoon and its inlets, with third-generation physics, depth-induced breaking (Battjes and Janssen, 1978) and JONSWAP bottom friction. Flow and waves exchanged fields every 3600 s through the DIMR framework, an interval set by the tidal and sea-breeze time scales the coupling has to resolve, and wave forces entered the flow as three-dimensional dissipation. Within each exchange the wave model ran in stationary mode, which holds because wave energy crosses the 2.5 km basin in about fifteen minutes. The spectral discretisation, the source terms and the ground for each setting are given in Section S2 and Table S1.

3.3. Boundary and atmospheric forcing

Open-sea boundaries were forced with the CMEMS Mediterranean Sea physical reanalysis. Water level, salinity, and temperature were prescribed along the offshore boundary, together with a depth-resolved advection velocity, which was necessary for stability in the deep offshore cells, where a water-level-only boundary admitted spurious accelerations. Since the CMEMS sea surface height is an anomaly about the model rest state rather than a geodetic level,

a per-node offset of +0.449 m, anchored on the Marettimo tide gauge over 384 days and distributed along the boundary by the CMEMS mean dynamic topography, was added to the water-level boundary. Its seasonal residual of about 0.07 m bounds the accuracy of the reference level the model can be held to (Section S3).

Wind components, mean sea level pressure and the Charnock parameter were taken from ERA5, the wind field blended with the record of the AltaVilaEst station, and evaporation was imposed as negative rainfall. Interior salinity and temperature were nudged towards the CMEMS fields with a relaxation time of 8.64×10^5 s, long enough to leave the lagoon interior essentially free-running while constraining the offshore extension.

Temperature and salinity outside the lagoon were initialised with the depth profiles of the CMEMS nudging field at the start time, applied over the domain, from 25.8 °C and 37.5 at the surface to 16.2 °C and 38.3 below 260 m. The hypersaline value of 42 was prescribed over the lagoon, whose temperature takes the surface value of the profile.

3.4. Seagrass canopy parametrisation

The canopy was represented as a three-dimensional drag field, resolved per sigma layer over the vertical extent the plants occupy. A cell whose canopy fills two fifths of the water column carries drag in the layers within that fraction and none above it. Drag per unit volume follows the standard quadratic form (Nepf, 2012), with a drag coefficient C_D, a stem diameter D, an areal stem density m and a canopy height h_v.

The alternative available in this solver is the trachytope framework, in which vegetation enters as a composite bed roughness following Baptist et al. (2007). It was not adopted here, since a roughness term acts at the bed whereas the quantity the factorial has to separate is a momentum sink distributed through the part of the column the plants occupy. For the same reason the calibrated value of Ingrassia et al. (2024) for this lagoon, a Gauckler–Strickler coefficient of 20 $\mathrm{m}^{1/3}\ \mathrm{s}^{-1}$ equivalent to a Manning coefficient of 0.050 $\mathrm{s\ m}^{-1/3}$, is a bulk roughness of the vegetated basin, and used as the bare-substrate roughness of a model that resolves the canopy it would count the canopy twice.

Table 1 gives the values and their provenance. None was fitted to any observation used to evaluate the model. Densities are expressed per leaf, since it is the leaf that presents the frontal area, and the canopy height is the deflected extent of the drag rather than the

leaf length, taken within the range of the flume and large-scale measurements of the species (Section S4). Fractional cover from the classification multiplies the areal density and nothing else, height and diameter being properties of the plant rather than of how much bed it covers, and median cover is 0.685 for *P. oceanica* and 0.181 for *C. nodosa*, which therefore carries about 1.5% of the total canopy frontal area.

Table 1: Canopy parameters. *P. oceanica* values are from Ciraolo et al. (2006b), except the canopy height, which is from Prinos et al. (2010). *C. nodosa* values are from Morris et al. (2008), with the shoot density also bracketed by Cancemi et al. (2002).

Parameter	*P. oceanica*	*C. nodosa*
Shoot density (m^{-2})	500	800
Leaves per shoot	6	3
Leaf width D (m)	0.010	0.004
Canopy height h_v (m)	0.40	0.15
Leaf density (kg m^{-3})	900	900
Drag coefficient C_D	0.8	0.8

The spatial assignment of classes came from a supervised random-forest classification of PlanetScope SuperDove imagery acquired in August 2023 at 3 m ground sampling distance, following the depth-invariant approach of Maltese et al. (2025), and classified pixels were transferred to the flow links of the mesh, a link straddling more than one class carrying each weighted by the area it occupies (Figure 2, Section S4).

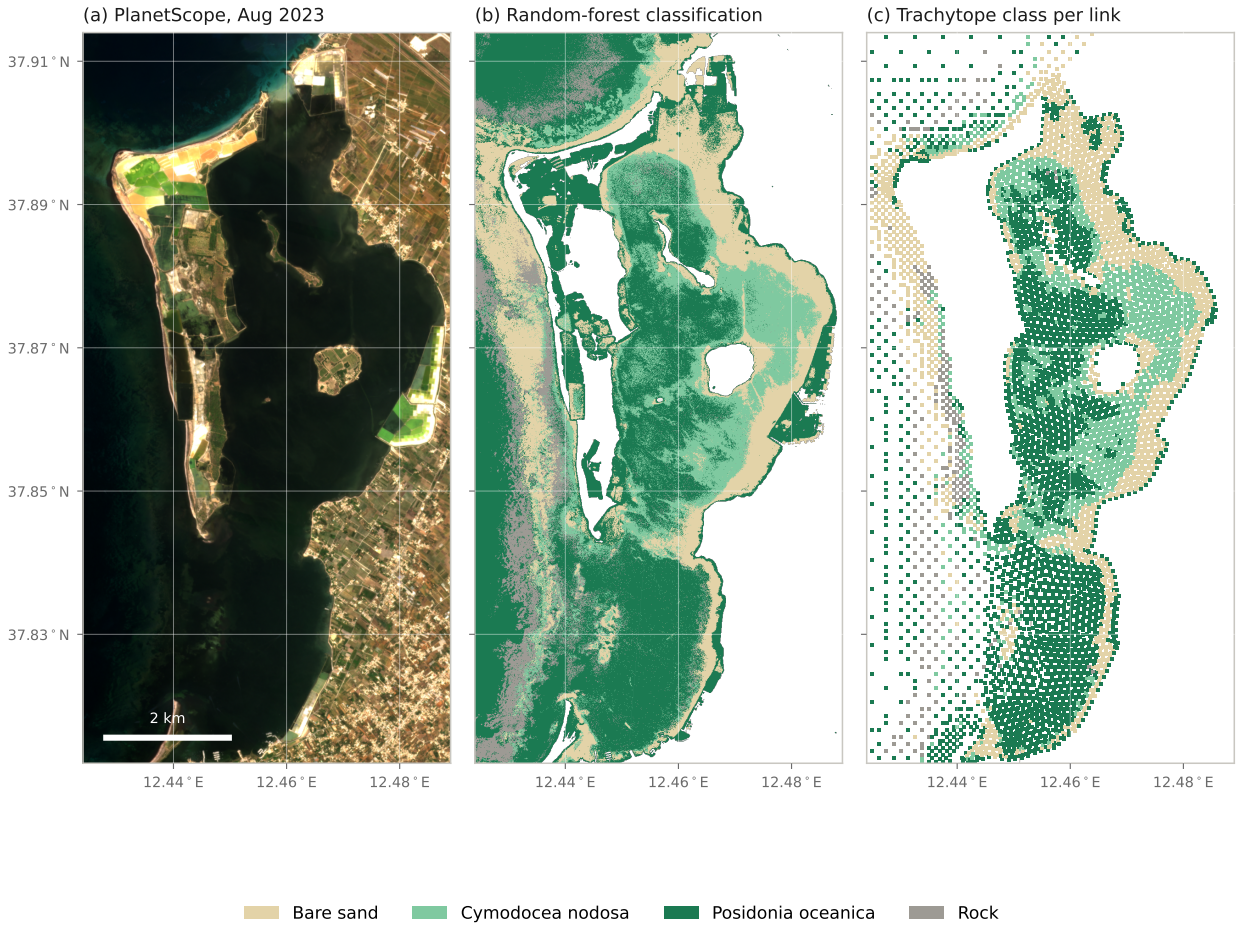


Figure 2: The seagrass map from imagery to model input. (a) PlanetScope SuperDove August 2023 median composite. (b) The random-forest classification, with the six output classes collapsed to the four seagrass classes. (c) The dominant class at each flow link, from the classification file the model reads, showing the mesh resolution the classification is sampled onto. The classification is computed over the whole scene, but only links inside the lagoon carry a canopy class.

3.5. Observational data and validation metrics

Water level was recorded at ten-minute resolution at three stations, BocaNord at the northern inlet, BocaSud near the southern inlet, and AltaVilaEst on the eastern flank of the central basin. Offshore performance was assessed at the Marettimo tide gauge.

Water-level skill was quantified with the raw root-mean-square error, the bias, and the anomaly root-mean-square error computed after removing the respective means, together with the linear correlation and the ratio of modelled to observed standard deviation. The three error measures are related by $\mathrm{RMSE}^2 = \mathrm{bias}^2 + \mathrm{RMSE}^2_{\mathrm{anom}}$, so reporting them jointly separates a constant reference-level offset from genuine dynamic error in phase and amplitude.

The first simulated day was discarded as spin-up, over which the initial condition relaxes and the transients it generates leave the basin.

Lagrangian skill was assessed against GPS-tracked surface drifters released in the lagoon interior during the simulation window. Simulated trajectories were obtained by advecting particles with OpenDrift (Dagestad et al., 2018) through the modelled surface current field, at an advection step of 300 s and with positions written every 600 s. Scoring was restricted, for each drifter, to the interval between its first and last observation, since the advection continues to the end of the forcing while the deployments last between 0.4 and 7.2 h. A wind drift factor of 0.02 was applied to every member, since a sweep over this parameter found no single value optimal across deployments and a per-member value would have confounded the process attribution with a tuning difference.

Agreement was quantified with the normalised cumulative Lagrangian separation skill score of Liu and Weisberg (2011), which accumulates the separation between modelled and observed positions and normalises it by the distance travelled by the observed drifter, and with the endpoint proximity, the separation between modelled and observed positions at the end of each deployment. Révelard et al. (2021) discuss the sensitivity of this class of metric to the normalisation choice in coastal applications.

3.6. Numerical experiments

Eight simulations were run over the same nine-day window, from the same initial condition and with identical boundary and atmospheric forcing, so that the differences between them isolate individual model components. All eight share the mesh, and all wrote station output every 600 s and gridded output every 1800 s, so the comparison is not affected by differences in output sampling.

Members are identified by a three-slot code whose slots mark wave coupling, seagrass canopy and mobile bed in that order, carrying `W`, `V` or `M` where the factor is active and a dash where it is not. The eight are `---`, `-V-`, `W--`, `WV-`, `--M`, `-VM`, `W-M` and `WVM`, and the code is used throughout the results.

The design is a complete factorial in three factors, wave coupling, seagrass canopy and bed mobility, and all eight cells were obtained. The process attribution is made on the four

fixed-bed members, a two-factor design in wave coupling and canopy, and the four mobile-bed members form a sensitivity arm, the critical shear stress for erosion not having been constrained against a repeated bathymetric survey. Each vegetated member is a three-day segment restarted on 7 July from the state of the bare member of the same cell, so the two members of a pair share the first six days and differ by the canopy over the window in which they are scored. In the mobile-bed members suspended sediment does not contribute to the density field, and erodible sediment is confined to depths above 15 to 20 m, where waves can plausibly move it (Section S5). The effect of the mobile bed on transport was traced with two segments of the wave-coupled mobile-bed member without canopy, restarted on 7 July from its state on that date, one with the bed updated as in the member and one with the bed held at that state.

The whole ensemble was also repeated from a vertically uniform column, at 24 °C and 37.5 outside the lagoon and with the same hypersaline value inside it. The horizontal and vertical resolution, a time-step limit by the vertical fluxes, and the host and decomposition were tested on the uncoupled fixed-bed configuration without canopy, as described in Section S6.

4. Results

All eight members are scored on a common interval, 8 to 10 July 2025, which is the intersection of the members' post-spinup spans and which contains the whole drifter record. Members are coded by wave coupling (W), seagrass canopy (V) and mobile bed (M). The attribution rests on the four fixed-bed members, and the mobile-bed arm is reported beside them in the tables and examined in Section 4.6.

4.1. Water level

Table 2 reports water level skill at the three interior stations. Raw error is dominated by a positive bias at BocaNord and AltaVilaEst, between 0.03 and 0.09 m, and by a negative one at BocaSud, between −0.051 and −0.015 m. The bias is a datum and boundary-offset property that all members share and it is not what distinguishes them.

The anomaly RMSE responds to different factors at the three stations (Figure 3b). At BocaSud it varies by 1.6 mm across the four fixed-bed members. At BocaNord it varies by

12 mm and follows wave coupling, falling from 0.039 to 0.027 m without a canopy and from 0.038 to 0.031 m with one. At AltaVilaEst, the innermost station, in the northern basin where the meadow is densest, it varies by 32 mm and follows the canopy, the two vegetated members giving 0.021 and 0.028 m with correlations of 0.94 and 0.95 and the two bare members 0.048 and 0.054 m with correlations of 0.79 and 0.81. Modelled-to-observed standard deviation ratios move the same way, from 1.18 and 1.37 without a canopy to 1.01 and 1.18 with one. The mobile-bed members reproduce the values of their fixed-bed counterparts to within 3 mm, with one exception, the bare wave-coupled member at AltaVilaEst, whose anomaly RMSE is 8 mm lower than on the fixed bed and remains 16 mm above that of its vegetated counterpart.

The record itself shows where that error sits (Figure 3a), with the bare configurations overshooting the range of the observed anomaly and running ahead of it, most visibly on the falling limb of 9 July, while the vegetated ones track both the amplitude and the phase. The two arms are separated by more than the spread within either of them over most of the window.

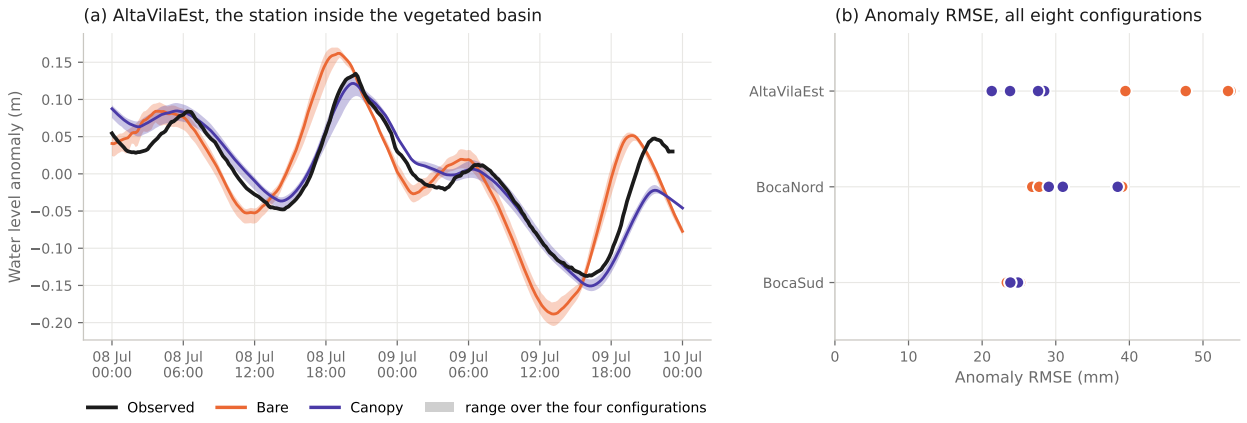


Figure 3: Water level over the scored window. (a) Anomaly at AltaVilaEst, observed against the two arms, each drawn as its mean and the envelope of its four configurations, the mobile-bed ones included. Anomalies rather than raw levels, because the bias is common to all eight members. (b) Anomaly RMSE at the three stations for every configuration, coloured by canopy treatment.

The vegetated members are three-day segments restarted from the bare member of the same cell, and a segment run without any other change reproduces its continuous member in water level and drifter skill (Supplementary Material, Section S7), so the separation at AltaVilaEst does not reflect the restart.

Table 2: Water level skill by member and station, 8–10 July 2025. $\mathrm{RMSE}^2 = \mathrm{bias}^2 + \mathrm{RMSE}^2_{\mathrm{anom}}$. Members are coded by wave coupling (W), seagrass canopy (V) and mobile bed (M); the last four of each station form the mobile-bed arm. All values in metres except the dimensionless ratios.

Station	Member	RMSE	Bias	$\mathrm{RMSE}_{\mathrm{anom}}$	r	σ_m/σ_o
BocaNord	`---`	0.052	+0.035	0.039	0.89	1.20
	`-V-`	0.050	+0.033	0.038	0.90	1.22
	`W--`	0.065	+0.059	0.027	0.94	1.08
	`WV-`	0.059	+0.051	0.031	0.92	1.09
	`--M`	0.052	+0.034	0.039	0.89	1.20
	`-VM`	0.050	+0.032	0.038	0.90	1.22
	`W-M`	0.066	+0.060	0.028	0.94	1.12
	`WVM`	0.059	+0.051	0.029	0.94	1.15
BocaSud	`---`	0.050	−0.043	0.025	0.95	1.20
	`-V-`	0.057	−0.051	0.025	0.95	1.16
	`W--`	0.028	−0.015	0.024	0.94	1.06
	`WV-`	0.035	−0.025	0.025	0.93	1.02
	`--M`	0.049	−0.043	0.025	0.95	1.20
	`-VM`	0.057	−0.051	0.025	0.95	1.16
	`W-M`	0.033	−0.023	0.023	0.94	1.08
	`WVM`	0.037	−0.028	0.024	0.93	1.05
AltaVila Est	`---`	0.076	+0.053	0.054	0.81	1.37
	`-V-`	0.049	+0.040	0.028	0.94	1.18
	`W--`	0.097	+0.085	0.048	0.79	1.18
	`WV-`	0.070	+0.067	0.021	0.95	1.01
	`--M`	0.077	+0.055	0.053	0.81	1.36
	`-VM`	0.050	+0.042	0.028	0.94	1.17
	`W-M`	0.074	+0.062	0.040	0.86	1.19
	`WVM`	0.062	+0.057	0.024	0.94	1.01

4.2. Lagrangian transport

Thirty-five drifters across twelve deployments were scored against each member over the interval each drifter was in the water, which ranged from 0.4 to 7.2 hours. Table 3 summarises the metrics and Figure 4 shows their distribution.

On the fixed bed, mean Liu–Weisberg skill spans 0.597 to 0.739 and endpoint separation 327 to 481 m, and the four members divide cleanly on the canopy. The two vegetated members score 0.711 and 0.739 and the two bare ones 0.597 and 0.619, and the two pairs do not overlap. The two components of the trajectory error move together and both improve with the canopy, path length ratio rising from 0.80 and 0.83 without a canopy to 0.88 and 0.94 with one, and

mean heading error falling from 11.2 and 10.4° to 8.5 and 8.4°.

Table 3: Lagrangian metrics by ensemble member, 35 drifters over twelve deployments, scored on the interval each drifter was in the water. Path ratio is simulated over observed path length. Heading error is the mean angular difference between simulated and observed step directions and is computed on eleven deployments, deployment 4 having too few positions to form the three step bearings it requires. The lower block is the mobile-bed arm (Section 4.6).

Member	Waves	Canopy	LW skill	Endpoint (m)	Path	Heading
`---`	no	no	0.619	478	0.80	11.2°
`-V-`	no	yes	**0.739**	**327**	0.88	8.5°
`W--`	yes	no	0.597	481	0.83	10.4°
`WV-`	yes	yes	0.711	335	**0.94**	**8.4°**
Mobile bed						
`--M`	no	no	0.614	479	0.80	11.4°
`-VM`	no	yes	0.739	330	0.88	8.5°
`W-M`	yes	no	0.708	304	0.89	7.9°
`WVM`	yes	yes	0.718	307	0.88	8.0°

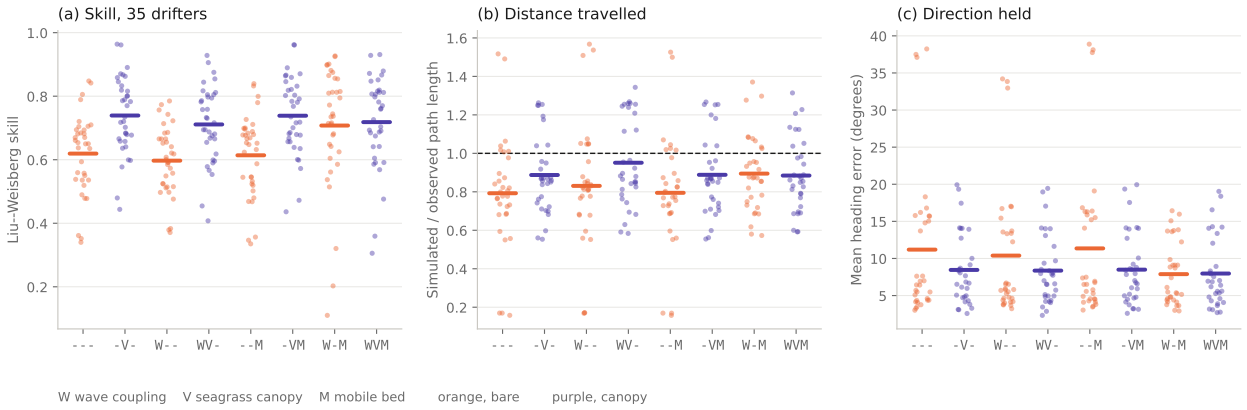


Figure 4: Lagrangian validation, one point per drifter and the bar giving the mean, the four fixed-bed members on the left and the mobile-bed arm on the right. (a) Liu–Weisberg skill. (b) Simulated over observed path length, the dashed line marking equality. (c) Mean angular difference between simulated and observed step directions. Panels (b) and (c) carry 32 drifters rather than 35, deployment 4 having too few positions to form step bearings.

Figure 5 shows the tracks for every deployment and for the wave-coupled fixed-bed pair that differs only in the canopy. In the central basin the bare particles leave the observed path early and finish short of the observed endpoint while the vegetated ones hold its heading, and in the southern deployments both follow the observed path closely, deployment 4 being the exception.

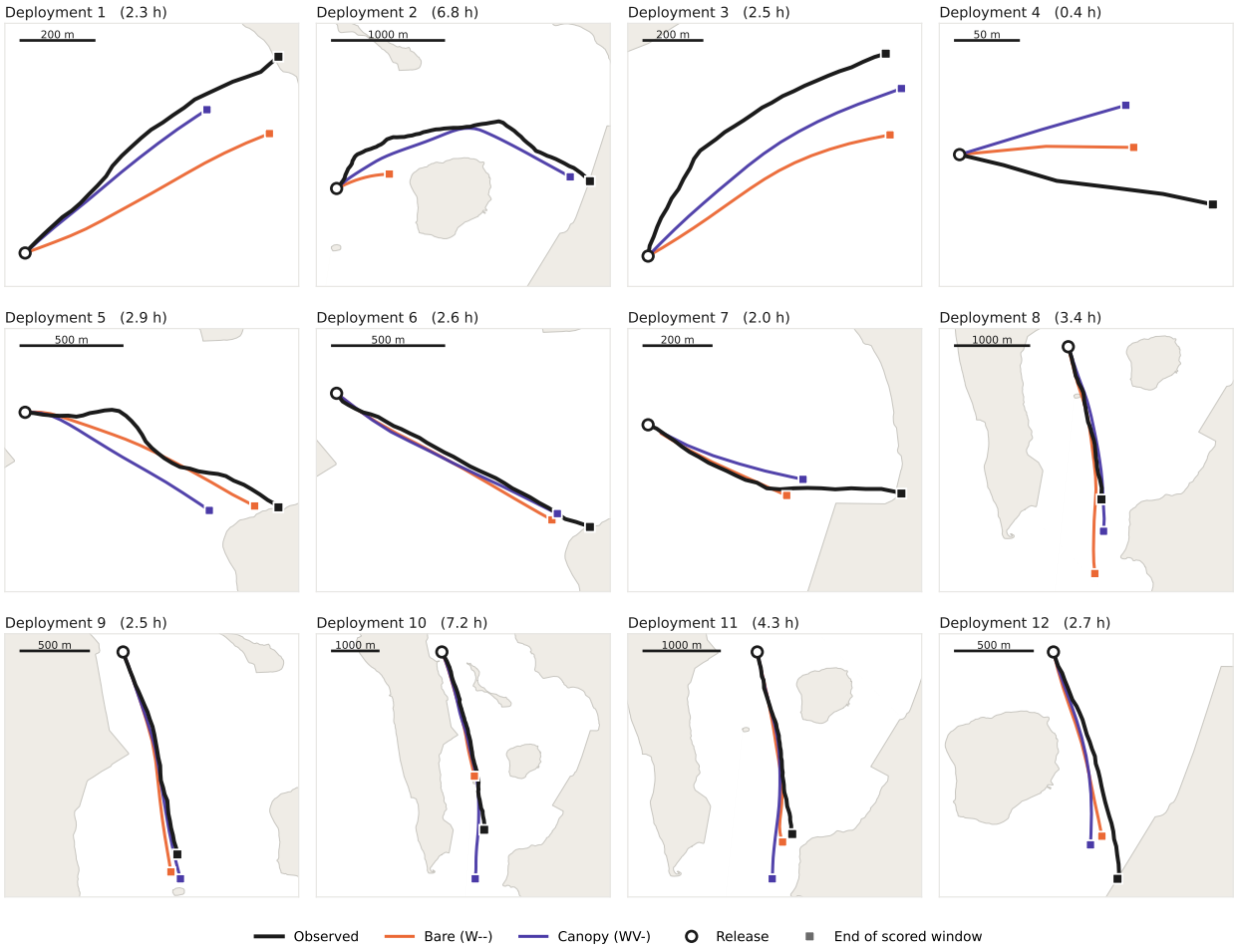


Figure 5: Observed and simulated drifter trajectories, all twelve deployments, for the wave-coupled fixed-bed configurations that differ only in the canopy. One drifter per panel, the median-skill drifter of its deployment under the vegetated configuration. Circles mark the release and squares the end of the scored window. Each track is clipped to the interval its drifter was in the water. Panel extents differ, so each carries its own scale bar.

4.3. Process attribution

On the fixed bed the design is a two-factor factorial in wave coupling and canopy, with two single-factor contrasts per factor. Figure 6 gives them, with the contrasts that involve the mobile bed beside them, with bootstrap confidence intervals over 4000 paired resamples.

Canopy. The canopy raises skill by 0.120 (95% CI +0.061 to +0.180) without waves and by 0.114 (CI +0.056 to +0.170) with them, with 25 and 27 of the 35 drifters improving. Both intervals are clear of zero, and the two contrasts agree to within 0.006, so the canopy effect does not depend on the wave treatment.

Wave coupling. Wave coupling reduces skill by 0.022 (CI −0.043 to −0.003) without

a canopy and by 0.028 (CI −0.040 to −0.017) with one, with 21 and 25 of the 35 drifters worse. The effect is a fifth of that of the canopy, and the interaction term of the two factors is −0.006.

The canopy contrast holds deployment by deployment (Figure 6b), with the same sign under both wave treatments in ten of the twelve deployments. Deployment 4, the shortest record and the one deployment in which the bare members score higher, is retained, and excluding it widens the two canopy contrasts to +0.148 and +0.144 (Section S7).

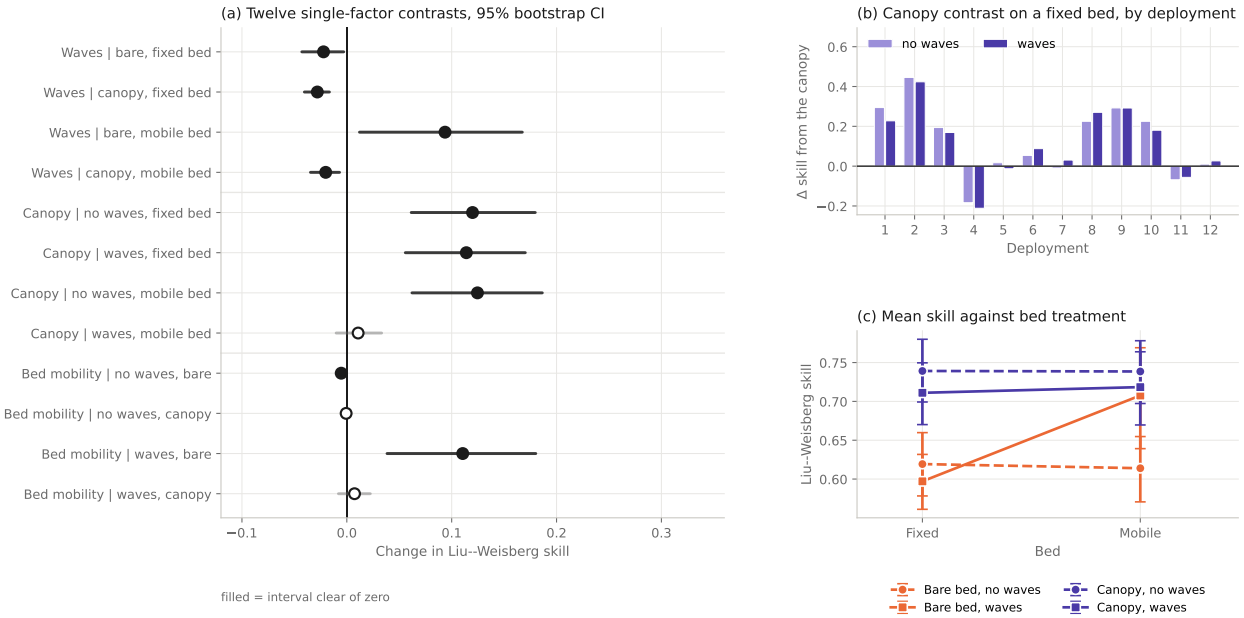


Figure 6: Process attribution. (a) The twelve single-factor contrasts of the three-factor design with 95% bootstrap intervals over 4000 paired resamples, filled markers denoting intervals clear of zero; the contrasts involving the mobile bed belong to the sensitivity arm. (b) The canopy contrast on the fixed bed, deployment by deployment, without and with wave coupling. (c) Mean skill against bed treatment, one line per combination of canopy and wave treatment, with 95% intervals on each mean.

4.4. Three-dimensional structure

Over the wet cells of the lagoon interior, with the inlets excluded, mean current speed increases monotonically from the bed to the surface in every member (Figure 7), and the canopy sets the shape of that profile. On the fixed bed without a canopy, bed-layer speed is 0.032 to 0.035 $\mathrm{m\,s^{-1}}$ and surface-layer speed 0.065 to 0.069 $\mathrm{m\,s^{-1}}$, for surface-to-bed ratios of 1.88 and 2.14. With a canopy, bed-layer speed falls to 0.0093 to 0.0095 $\mathrm{m\,s^{-1}}$ while surface speed falls much less, to 0.043 to 0.047 $\mathrm{m\,s^{-1}}$, and the ratio rises to 4.50 and 5.05.

Wave coupling moves the ratio by far less, adding 0.26 in the bare arm and 0.55 in the vegetated one, against a canopy contribution of +2.62 without waves and +2.91 with them.

Bed shear stress separates the two treatments by sign rather than by magnitude. In the one-day probes used to set the canopy parameters, started from the uniform column, it fell over the meadow from 0.037 to 0.0026 Pa when the canopy was introduced, a factor of 14, while a probe representing the same meadow as an equivalent bed roughness instead of as a canopy raised it to 0.049 Pa over the same cells and times.

Lagoon-mean surface speed does not order the members the way transport does, since the member reproducing the observed drifter behaviour best has the lowest lagoon-mean surface speed of the fixed-bed members at 0.043 $\mathrm{m\,s^{-1}}$, against 0.069 for a member whose particles travel 17% too short a distance.

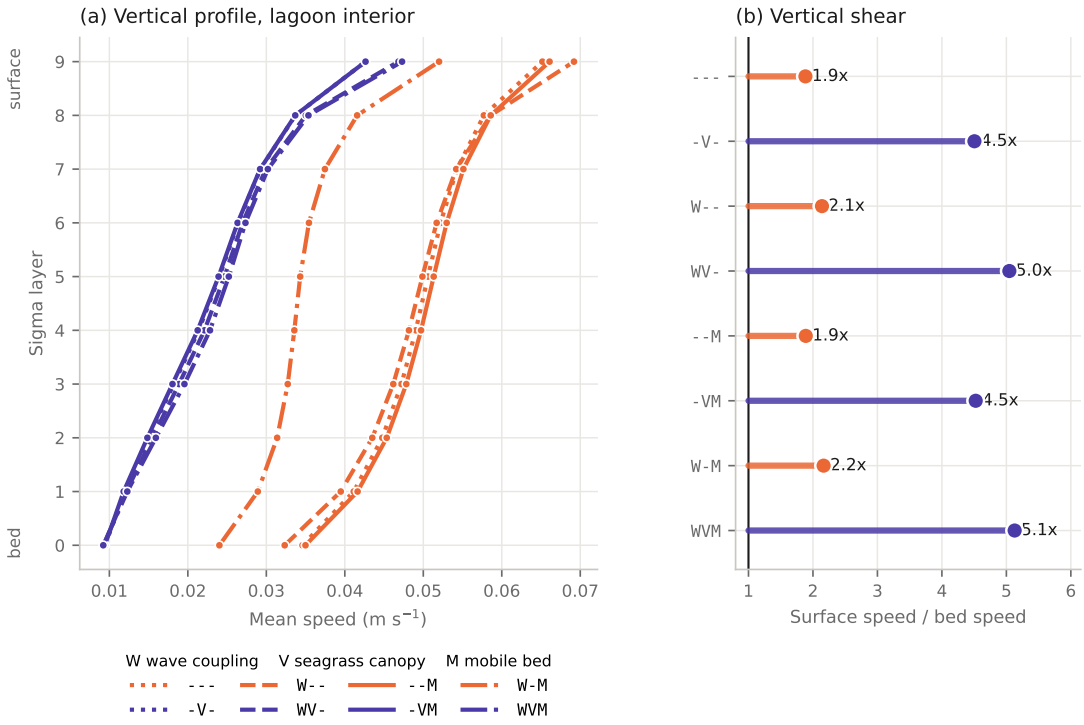


Figure 7: Vertical structure in the lagoon interior over the drifter window. (a) Mean speed by sigma layer, colour giving the canopy treatment and line style the wave and bed treatments. (b) The ratio of surface to bed speed.

4.5. Response of the seagrass classes

Cells were grouped by the seagrass class of the nearest classified point. The classes occupy different parts of the basin at different depths, and the bare fixed-bed configurations already

have their *Posidonia* cells slower at the bed than their sand cells, in a ratio of 0.91 and 0.92, so the canopy effect is read between configurations within a class or as a change in that ratio.

Under either reading the effect is large and correctly ordered, since on the fixed bed introducing the canopy reduces mean bed speed by 78% over *Posidonia* and 77% over *Cymodocea*, against 59% over sand and 57% over rock (Figure 8b). The *Posidonia* to sand bed-speed ratio falls from 0.91 and 0.92 without a canopy to 0.49 with one, under both wave treatments. The two unvegetated classes respond as well, because slowing the flow through the meadow changes the circulation of a basin whose interior is largely meadow, but they respond by a quarter less.

The canopy also removes most of the dependence on wave coupling, since within a class mean bed speed differs between the two bare fixed-bed configurations by 0.002 to 0.010 $\mathrm{m\,s^{-1}}$ and between the two vegetated ones by 0.0002 to 0.0011 $\mathrm{m\,s^{-1}}$. Once the canopy is present the near-bed flow is nearly the same whatever the wave treatment, which is the near-bed counterpart of the agreement in transport skill reported in Section 4.2.

The wave field does not follow the flow, mean orbital velocity at the bed changing by 0.2 to 0.9% over the four classes when the canopy is introduced on the fixed bed (Figure 8c), since the canopy does not reach the wave computation (Section 5.4).

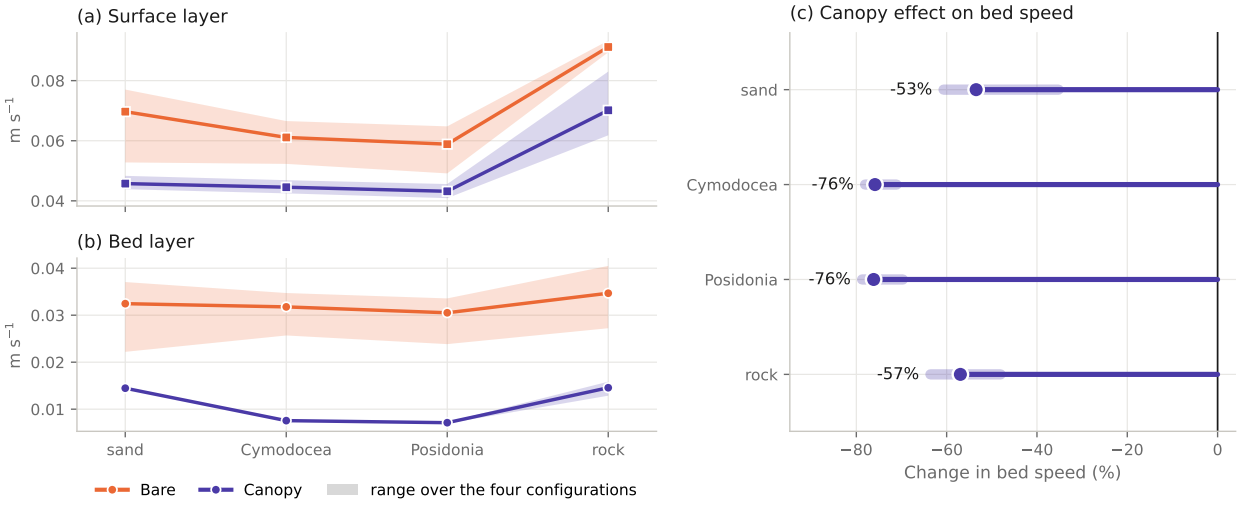


Figure 8: Flow response by seagrass class in the lagoon interior. (a) Mean speed in the surface layer and (b) in the bed layer, the line giving the mean over the four configurations of each arm, the mobile-bed ones included, and the band their range. (c) Change in mean bed speed on introducing the canopy, the marker giving the mean over the four cells of the design and the band their range.

4.6. Sensitivity to bed mobility, initial condition and discretisation

With wave coupling the mobile bed leaves any plausible state of the basin. Averaged over the lagoon from 7 to 10 July, suspended sediment reaches 9.3 $\mathrm{g\,L^{-1}}$ in the bare wave-coupled member and 9.4 $\mathrm{g\,L^{-1}}$ in its vegetated counterpart, against 0.1 and 0.007 $\mathrm{g\,L^{-1}}$ in the two uncoupled mobile-bed members, and the bed of the lagoon loses 4.0 and 3.1 million $\mathrm{m^3}$ over the nine days. By 10 July the bare wave-coupled member has lowered the lagoon bed by 12 cm on average and changed it by more than 5 cm over 74% of its area, whereas without waves the change exceeds 5 cm over 0.3% of the area. The canopy does not shield the bed from this erosion, because it does not reach the wave model.

The same bare member nevertheless scores 0.708 (Table 3), close to the vegetated members, so that in the cell combining waves and a mobile bed the canopy contrast falls to +0.011 (CI −0.010 to +0.033) and the wave and bed-mobility contrasts of the bare arm reach +0.094 and +0.111 (Figure 6a). The two segments restarted on 7 July locate the source of that skill. With the bed updated the segment scores 0.710 and with the bed held at its 7 July state 0.707, a difference of -0.003 ± 0.004, while the held bed scores 0.110 ± 0.039 above the fixed-bed member `W--`, with 29 of 35 drifters better. The gain therefore comes from the bathymetry reshaped over the first six days and not from sediment transport in the drifter window. At AltaVilaEst the member keeps an anomaly RMSE of 0.040 m and a correlation of 0.86, against 0.024 m and 0.94 for its vegetated counterpart.

Started from a vertically uniform column instead of the stratified state, the bare members score 0.080 to 0.145 lower and the vegetated members 0.005 to 0.018 lower, so that the canopy contrasts on the fixed bed become +0.195 and +0.188. The uniform start raises the modelled mean water level at the gauges by 2 to 10 cm and, on the fixed bed, changes anomaly RMSE by at most 4 mm and correlation by at most 0.03, and the canopy signature at AltaVilaEst is the same under both starts.

The horizontal and vertical resolution, the time-step limit and the host and decomposition change skill by less than 0.01 under a Courant limit of 0.5 and water level by at most 2.4 mm, so they do not limit the quantities the study reports (Section S6). Under the Courant limit of 0.7 used by the ensemble the uncoupled member scores 0.034 ± 0.004 lower on four partitions on a second computer than on eight on the server, a dependence that all members share.

5. Discussion

5.1. Vertical placement of the canopy drag

This study asked which modelled processes control transport in a shallow vegetated lagoon, and on the fixed bed the answer the ensemble gives is the seagrass canopy, with an effect five times that of wave coupling and the same under both wave treatments. The mechanism the model offers for that is vertical rather than frictional, since introducing the canopy removes momentum from the lower part of the water column without removing a proportionate amount from the upper part, more than doubling the ratio of surface to bed speed. Such a profile is the signature of a canopy shear layer rather than of a rougher bed, which acts through a logarithmic profile whose surface-to-bed ratio is bounded well below the value obtained here (Nepf, 2012), and the bed shear stress confirms the distinction by its sign, falling under the canopy where an equivalent bed roughness raises it over the same cells. Since a surface drifter samples the top of that profile, a treatment that reshapes the profile relocates the drifter even where it changes the depth-averaged transport little.

The result stands in a complementary relation to the two earlier models of this basin. In the depth-averaged framework of Ingrassia et al. (2024) an equivalent bed roughness is the appropriate representation of a canopy, since there is no vertical profile for the canopy to shape, and the three-dimensional circulation established by De Marchis et al. (2012) is what motivates a three-dimensional solver here. The contribution of the present work is to quantify what a canopy formulation resolving the drag over the vertical extent of the plants adds in a basin of this depth. Its parameters were taken from published measurement and not adjusted against any observation, so the improvement follows from where the drag is applied rather than from how it was tuned.

5.2. Information carried by each gauge position

Water level separated the configurations by the canopy at AltaVilaEst, by wave coupling at BocaNord and by neither at BocaSud, and the three positions constrain different parts of the problem. Agreement at the inlet gauges establishes that the offshore forcing, the datum and the exchange through the two mouths are right, and in a micro-tidal basin with two open connections those set most of the level there, the wave contribution to the northern inlet

being the one dynamic effect that reaches it. Agreement in the interior tests something the outer pair cannot, because the level there is set by what the water does on its way in, which is where the canopy acts. The roles are complementary, and a match in the interior would be difficult to interpret without the exchange having been established first.

Together with the drifter result, the gauge result refines the familiar argument that water level constrains transport weakly and that Lagrangian observations form a distinct constraint rather than a confirmation (Liu and Weisberg, 2011; Révelard et al., 2021). Water level is not inherently insensitive to the canopy, since the one gauge sited inside the vegetated basin identified the same factor as the drifters and ranked the fixed-bed configurations the same way. For a basin of this type the useful question is therefore less whether to validate on water level than where to place the instrument intended to discriminate, and the interior of the vegetated compartment is a considerably more informative position for that purpose than the inlet. Section 5.3 shows that the interior gauge also discriminates where the drifters do not.

5.3. Bare-bed skill through a reshaped bathymetry

The mobile-bed arm contains one configuration that, taken on its own, supports a strong statement about process coupling. In the bare arm, wave coupling and bed mobility each leave Lagrangian skill nearly unchanged or lower it when applied alone and raise it when applied together, with an interaction term of +0.116, tight bootstrap intervals and a mechanism available in the literature, since wave orbital motion sets the bed stress that mobilises sediment and wave–tide coupling is known to govern the sediment dynamics of shallow micro-tidal basins (Carniello et al., 2011). The combination brings the bare member to 0.708, within 0.03 of the vegetated members, and removes the canopy contrast in that cell.

The segments restarted on 7 July show that the statement has no physical basis in this model. The skill is carried by the bathymetry that six days of implausible wave-driven erosion reshaped (Section 4.6), a state the basin does not reach in summer. The reshaped bed does not reproduce the vertical structure of the canopy either, the surface-to-bed speed ratio of that member remaining at 2.17 against 5.13 with the canopy, so the agreement in skill is reached through a different flow. A model without a canopy can therefore reach the drifter skill of a model with one through a mechanism that is wrong, and the drifters alone do not

separate the two. The interior gauge does, keeping the bare member well apart from its vegetated counterpart.

The initial condition shows the same asymmetry by another route. A start from a uniform offshore column lowers the skill of the bare members by 0.08 to 0.15 and that of the vegetated members by at most 0.02, so the canopy contrast grows from about 0.12 to about 0.19 when the start is wrong. In both cases an error elsewhere in the model degrades the bare configurations and leaves the vegetated ones nearly unaffected, and the magnitude of a canopy contrast therefore depends on what else the model gets wrong, even where its sign does not.

What follows from this bears on how such an ensemble is read rather than on this basin alone, since canopy drag is commonly left out of lagoon and estuary models or folded into a bed roughness coefficient, and an ensemble built that way can still return a well-defined and statistically clean interaction between the processes it does carry. Here that interaction is produced by an implausible morphodynamic state, and nothing internal to the bare half of the design distinguishes it from a physical result. What distinguished them was varying the canopy as a factor in its own right, checking the morphodynamic state against a plausible sediment budget, and scoring the result against two observation types that no model decision had used.

5.4. Limitations

Wave dissipation by the canopy. The canopy does not reach the wave model, because the coupling passes a bed roughness field to the spectral model while the canopy enters the momentum equation of the flow, so mean orbital velocity at the bed changes by under 1% when the canopy is introduced. Seagrass meadows do attenuate waves, in this species and at this scale (Infantes et al., 2012; Prinos et al., 2010), and shallow restricted Mediterranean lagoons carry wave fields that respond to bed conditions (Morote-Sánchez et al., 2025). The wave field here is therefore that of a bare bed, and the wave contrasts, together with the wave-driven erosion of the mobile-bed arm, are conditional on that.

Canopy reconfiguration. The canopy bends under load, the velocity ratio between two probes differing only in leaf stiffness moving from 1.005 in the slowest quintile of the record to 0.807 in the fastest 5%, and a 13% change in leaf density moves the velocity field by 20% at the upper end of the range, a strong sensitivity to a value taken from a published

table rather than measured in this meadow. The bending is mechanical, whereas the flume measurements of Ciraolo et al. (2006b) describe reconfiguration empirically through a friction factor, a formulation available in the model only in its depth-averaged solver, so the agreement between the two was not tested.

Bed mobility. The morphodynamic arm is not calibrated, the critical shear stress for erosion having been constrained by neither a repeated bathymetric survey nor a turbidity record, and with waves it leaves any plausible state of the basin, with suspended sediment near 9 $\mathrm{g\,L^{-1}}$ and the lagoon bed lowered by 12 cm on average in nine days (Section 4.6). It is therefore a sensitivity test and is kept out of the attribution. Before it could enter it, the erosion parameters would need to bring suspended sediment in the lagoon to the order of tens of milligrams per litre.

Window and restart segments. The record covers two days of water level and thirty-four hours of drifter observations within one summer season, and the vegetated configurations are three-day restart segments. A segment run without any other change reproduces its continuous member in water level and drifter skill (Section 4.1), but seasonal behaviour, storm conditions and the interaction of a canopy with a mobile bed over longer periods are outside what this design addresses.

Initial condition. The stratified start is a single profile applied over the domain and leaves out the horizontal structure of the CMEMS field, so the canopy contrasts reported here may still carry part of a start-up adjustment, whose size the uniform start bounds (Section 5.3). The solution also depends on the host and the decomposition at the Courant limit of the ensemble, by 0.034 in skill, which all members share.

***Cymodocea nodosa* parameters.** The *C. nodosa* parameters are loosely constrained, coming from a flume study of plants collected elsewhere, since no measurement of that species in this lagoon was available. The species carries about 1.5% of the canopy frontal area in the model, which bounds the consequence, and the drag is carried almost entirely by the *P. oceanica* column.

6. Conclusions

In a shallow Mediterranean lagoon where the seagrass canopy occupies most of the basin and the mean depth is below one metre, the canopy emerges from this ensemble as the first-order control on Lagrangian transport. On the fixed bed, representing it raises Liu–Weisberg skill by 0.120 without wave coupling and by 0.114 with it, both bootstrap intervals clear of zero, against -0.022 and -0.028 for wave coupling, and the two factors do not interact.

The results indicate that where the drag is applied matters more than how it is calibrated. The canopy here is a three-dimensional drag field parameterised entirely from published measurement of the two species, with nothing fitted, and it more than doubles the ratio of surface to bed speed while lowering bed shear stress by a factor of 14, where representing the same meadow as an equivalent bed roughness raises that stress. A treatment that changes the shape of the velocity profile relocates a surface drifter, where a treatment changing only its magnitude does not.

It follows that process attribution from a factorial is conditional on the factors it contains and on the state the model reaches. With waves, the uncalibrated mobile bed brings a configuration without a canopy to within 0.03 of the vegetated ones through a bathymetry reshaped by implausible erosion, and a uniform offshore start raises the canopy contrasts to 0.19 by degrading the bare configurations alone, two compensations that no statistic internal to the bare configurations reveals.

The same design shows that where a water level gauge sits decides what it can constrain. Anomaly RMSE differs across the four fixed-bed configurations by 1.6 mm at the southern inlet, by 12 mm at the northern inlet, where it follows wave coupling, and by 32 mm at the gauge inside the vegetated basin, where it follows the canopy and correlation moves from 0.79–0.81 to 0.94–0.95. The interior gauge also separates the canopy treatments in the configuration where the drifters do not, the bare wave-coupled mobile-bed member keeping an anomaly RMSE of 0.040 m against 0.024 m with the canopy.

The canopy is not a fixed property of this basin, its extent having fallen substantially over the past two decades, and the two arms of the design bracket what a loss of that kind does to the flow, one carrying the meadow as mapped and the other none of it. Between those two states mean bed speed over the meadow changes by 78%, the surface layer moves from

about twice the speed of the bed to about five times, and Lagrangian skill separates without overlap. The transport regime of a basin of this type is therefore a property of its meadow as much as of its geometry and its forcing.

CRediT authorship contribution statement

Martins Jr, C.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **De Marchis, M.:** Conceptualization, Supervision, Methodology, Writing – review & editing. **Ciraolo, G.:** Conceptualization, Supervision, Methodology, Writing – review & editing. **Nasello, C.:** Conceptualization, Supervision, Methodology, Writing – review & editing. **Maltese, A.:** Conceptualization, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Software and data availability

- **Model:** Delft3D Flexible Mesh Suite 2026.01 HMWQ (D-Flow FM kernel 1.2.184) coupled online with SWAN through DIMR.
- **Source code repository:** `https://github.com/cicero-martins/StagnoneDT`
- **Data availability:** ERA5 and CMEMS products are openly available from the Copernicus programme. PlanetScope imagery is under commercial licence, and scene identifiers are listed in the repository for traceability. In-situ water level, wind, and drifter records are archived in the repository.

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